



[Practice Challenge](#) [Session 1 Recap](#)

# Programming Fundamentals

## Selection 1 — Decisions with if and if-else

Programming Fundamentals Team

SETU

2026-09-08



# Outline

- |                                |                                  |                                    |
|--------------------------------|----------------------------------|------------------------------------|
| 1. Today                       | 8. = versus ==                   | 16. When We Need Two Outcomes      |
| 2. Warm-up: What happens next? | 9. Predict the Result            | 17. if-else                        |
| 3. Flow of Control             | 10. The if Statement             | 18. Pass or Fail                   |
| 4. A Real Decision             | 11. Indentation Matters          | 19. Boundary Values Matter         |
| 5. Boolean Values              | 12. Trace an if                  | 20. Strings in Conditions          |
| 6. Boolean Expressions         | 13. One-way Selection            | 21. Making String Input Friendlier |
| 7. Relational Operators        | 14. Mini Activity: Free Delivery |                                    |
|                                | 15. Activity Solution            |                                    |



# Outline

**22. Common Errors**

**23. Fix the Program**

**24. Mini-program:  
Ticket Price**

**25. Think Before You  
Code**

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# Outline

## 1. Today

2. Warm-up: What happens next?

3. Flow of Control

4. A Real Decision

5. Boolean Values

6. Boolean Expressions

7. Relational Operators

8. = versus ==

9. Predict the Result

10. The if Statement

11. Indentation Matters

12. Trace an if

13. One-way Selection

14. Mini Activity: Free Delivery

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16. When We Need Two Outcomes

17. if-else

18. Pass or Fail

19. Boundary Values Matter

20. Strings in Conditions

21. Making String Input Friendlier



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# 1. Today

By the end of this session you should be able to:

- Explain sequence and selection.
- Recognise Boolean conditions.
- Use relational operators correctly.
- Write `if` and `if-else` statements.
- Trace a program and predict which branch runs.
- Use indentation correctly in Python.



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## 2. Warm-up: What happens next?

```
name = input("Name: ")
print("Hello", name)
print("Welcome to SETU")
```



## 2. Warm-up: What happens next?

```
name = input("Name: ")  
print("Hello", name)  
print("Welcome to SETU")
```

Every statement runs, from top to bottom.

**This is sequence:** the default flow of execution.



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## 3. Flow of Control

|                  |                                                   |
|------------------|---------------------------------------------------|
| <b>Sequence</b>  | Statements run in order                           |
| <b>Selection</b> | Statements run only when a condition is satisfied |
| <b>Iteration</b> | Statements are repeated                           |



## 3. Flow of Control

|                  |                                                   |
|------------------|---------------------------------------------------|
| <b>Sequence</b>  | Statements run in order                           |
| <b>Selection</b> | Statements run only when a condition is satisfied |
| <b>Iteration</b> | Statements are repeated                           |

Today we move from programs that always do the same thing to programs that can **make decisions**.



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## 4. A Real Decision

A cinema might ask:

```
age = int(input("Enter your age: "))
```

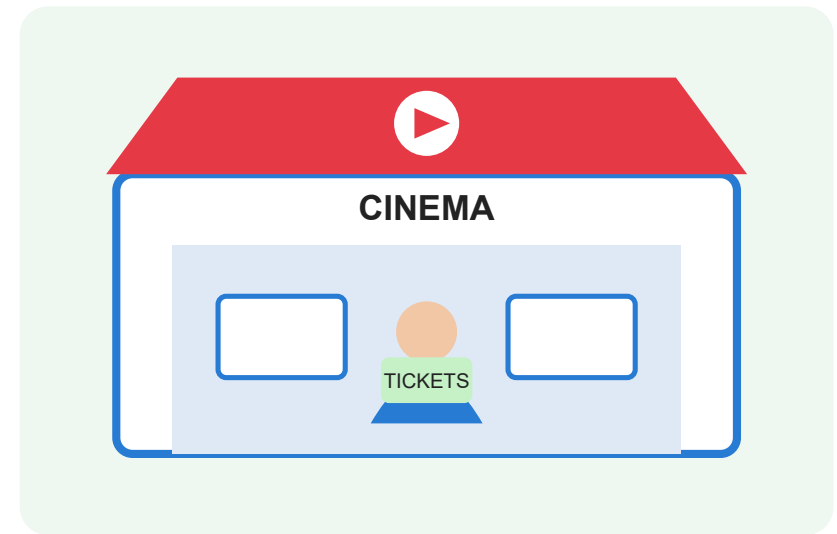
## 4. A Real Decision

A cinema might ask:

```
age = int(input("Enter your age: "))
```

- If age is 18 or more → allow entry.
- Otherwise → refuse entry.

The program needs a condition whose answer is **True** or **False**.





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## 5. Boolean Values

Python has a Boolean data type: `bool`.

```
is_raining = True
```

```
logged_in = False
```



## 5. Boolean Values

Python has a Boolean data type: `bool`.

```
is_raining = True
```

```
logged_in = False
```

There are only two Boolean values:

- `True`
- `False`



## 5. Boolean Values

Python has a Boolean data type: `bool`.

```
is_raining = True
```

```
logged_in = False
```

There are only two Boolean values:

- `True`
- `False`

Notice the capital letters.



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## 6. Boolean Expressions

A Boolean expression produces True or False.

```
age >= 18
```

```
mark >= 40
```

```
password == "python"
```

```
number < 0
```



## 6. Boolean Expressions

A Boolean expression produces True or False.

```
age >= 18
```

```
mark >= 40
```

```
password == "python"
```

```
number < 0
```

Try these in the Python REPL:

```
>>> 10 > 5
```

```
True
```

```
>>> 10 == 5
```

```
False
```



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## 7. Relational Operators

| Operator | Example                | Meaning                  |
|----------|------------------------|--------------------------|
| >        | <code>x &gt; y</code>  | greater than             |
| >=       | <code>x &gt;= y</code> | greater than or equal to |
| <        | <code>x &lt; y</code>  | less than                |
| <=       | <code>x &lt;= y</code> | less than or equal to    |
| ==       | <code>x == y</code>    | equal to                 |
| !=       | <code>x != y</code>    | not equal to             |



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## 8. = versus ==

age = 18

means **put 18 into age.**



## 8. = versus ==

age = 18

means **put 18 into age.**

age == 18

asks **is age equal to 18?**



## 8. = versus ==

age = 18

means **put 18 into age.**

age == 18

asks **is age equal to 18?**

**Common beginner error:** = assigns; == compares.



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## 9. Predict the Result

What does each expression produce?

`10 > 3`

`7 == 7`

`5 != 5`

`12 <= 10`

`"cat" == "cat"`



## 9. Predict the Result

What does each expression produce?

`10 > 3`

`7 == 7`

`5 != 5`

`12 <= 10`

`"cat" == "cat"`

Answers: True, True, False, False, True.



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## 10. The if Statement

```
if condition:  
    statement
```



## 10. The if Statement

```
if condition:  
    statement
```

Example:

```
age = 20
```

```
if age >= 18:  
    print("You are an adult")
```



## 10. The if Statement

```
if condition:  
    statement
```

Example:

```
age = 20
```

```
if age >= 18:  
    print("You are an adult")
```

The indented statement runs **only if** the condition is True.



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# 11. Indentation Matters

Correct:

```
if mark >= 40:
    print("Pass")
    print("Well done")
```



# 11. Indentation Matters

Correct:

```
if mark >= 40:
    print("Pass")
    print("Well done")
```

Incorrect:

```
if mark >= 40:
print("Pass")
```

Python uses indentation to define the block belonging to the if.



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## 12. Trace an if

```
temperature = 25
```

```
if temperature > 20:  
    print("Warm day")
```

```
print("Finished")
```



## 12. Trace an if

```
temperature = 25
```

```
if temperature > 20:  
    print("Warm day")
```

```
print("Finished")
```

Output:

Warm day  
Finished



## 12. Trace an if

```
temperature = 25
```

```
if temperature > 20:  
    print("Warm day")
```

```
print("Finished")
```

Output:

Warm day

Finished

What changes if temperature = 12?



# Outline

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## 13. One-way Selection

An if statement is **one-way selection**.

```
balance = 100
```

```
withdrawal = 40
```

```
if withdrawal <= balance:
```

```
    balance = balance - withdrawal
```

```
print(balance)
```



## 13. One-way Selection

An if statement is **one-way selection**.

```
balance = 100  
withdrawal = 40
```

```
if withdrawal <= balance:  
    balance = balance - withdrawal
```

```
print(balance)
```

If the condition is false, Python simply skips the indented block.



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## 14. Mini Activity: Free Delivery

An online shop gives free delivery when an order is €50 or more.

Complete:

```
order_total = float(input("Order total:  
€"))  
# your selection here
```



## 14. Mini Activity: Free Delivery

An online shop gives free delivery when an order is €50 or more.

Complete:

```
order_total = float(input("Order total:
€"))
# your selection here
```

Write the condition first. Then decide what belongs inside the block.





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## 15. Activity Solution

```
order_total = float(input("Order total: €"))

if order_total >= 50:
    print("Free delivery")
```



# 15. Activity Solution

```
order_total = float(input("Order total: €"))
```

```
if order_total >= 50:  
    print("Free delivery")
```

Question: what happens for an order of €30?



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## 16. When We Need Two Outcomes

Suppose every mark must be classified as either:

- Pass
- Fail



## 16. When We Need Two Outcomes

Suppose every mark must be classified as either:

- Pass
- Fail

We could write two separate if statements...

```
if mark >= 40:  
    print("Pass")  
if mark < 40:  
    print("Fail")
```

But Python gives us a clearer structure.



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## 17. if-else

```
if condition:
    statements_when_true
else:
    statements_when_false
```



## 17. if-else

```
if condition:
    statements_when_true
else:
    statements_when_false
```

Exactly **one** branch runs.



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## 18. Pass or Fail

```
mark = int(input("Enter mark: "))

if mark >= 40:
    print("Pass")
else:
    print("Fail")
```



## 18. Pass or Fail

```
mark = int(input("Enter mark: "))
```

```
if mark >= 40:  
    print("Pass")  
else:  
    print("Fail")
```

Trace it for:

- mark = 72
- mark = 38
- mark = 40



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## 19. Boundary Values Matter

Consider:

```
if age > 18:
    print("Adult")
else:
    print("Not adult")
```



## 19. Boundary Values Matter

Consider:

```
if age > 18:  
    print("Adult")  
else:  
    print("Not adult")
```

What happens when `age == 18`?



## 19. Boundary Values Matter

Consider:

```
if age > 18:  
    print("Adult")  
else:  
    print("Not adult")
```

What happens when `age == 18`?

If 18 counts as adult, the condition should be:

```
if age >= 18:
```

Always test values **at the boundary**.



# Outline

- |                                |                                  |                                    |
|--------------------------------|----------------------------------|------------------------------------|
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| 3. Flow of Control             | 10. The if Statement             | 18. Pass or Fail                   |
| 4. A Real Decision             | 11. Indentation Matters          | 19. Boundary Values Matter         |
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# Outline

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## 20. Strings in Conditions

Selection is not limited to numbers.

```
answer = input("Continue? y/n: ")
```

```
if answer == "y":  
    print("Continuing...")  
else:  
    print("Stopping...")
```



## 20. Strings in Conditions

Selection is not limited to numbers.

```
answer = input("Continue? y/n: ")
```

```
if answer == "y":  
    print("Continuing...")  
else:  
    print("Stopping...")
```

String comparisons are case-sensitive: "Y" != "y".



# Outline

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## 21. Making String Input Friendlier

```
answer = input("Continue? y/n: ").lower()

if answer == "y":
    print("Continuing...")
else:
    print("Stopping...")
```



## 21. Making String Input Friendlier

```
answer = input("Continue? y/n: ").lower()

if answer == "y":
    print("Continuing...")
else:
    print("Stopping...")
```

Now Y and y are both treated as "y".



# Outline

- |                                |                                  |                                    |
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# Outline

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## 22. Common Errors

Which errors can you spot?

```
age = input("Age: ")
```

```
if age = 18
    print("You are 18")
else
    print("You are not 18")
```



## 22. Common Errors

Which errors can you spot?

```
age = input("Age: ")
```

```
if age = 18
    print("You are 18")
else
    print("You are not 18")
```

Problems:

- `input()` returns a string.
- `=` should be `==` for comparison.
- `:` is missing.
- indentation is missing.



# Outline

- |                                |                                  |                                    |
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# Outline

22. Common Errors

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## 23. Fix the Program

Correct version:

```
age = int(input("Age: "))

if age == 18:
    print("You are 18")
else:
    print("You are not 18")
```



# Outline

- |                                |                                  |                                    |
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## 24. Mini-program: Ticket Price

Requirement:

- Under 16  $\rightarrow$  €8
- Everyone else  $\rightarrow$  €12

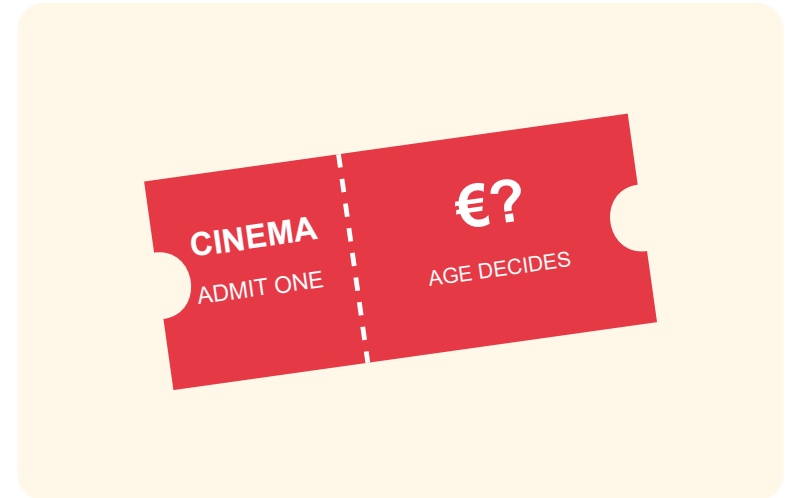


## 24. Mini-program: Ticket Price

Requirement:

- Under 16 → €8
- Everyone else → €12

```
age = int(input("Age: "))
if age < 16:
    price = 8
else:
    price = 12
print("Ticket price: €", price)
```





# Outline

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# Outline

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Ticket Price

**25. Think Before You  
Code**

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## 25. Think Before You Code

For each problem, identify:

1. What data do I need?
2. What is the Boolean condition?
3. What happens when it is true?
4. Do I need an else?



## 25. Think Before You Code

For each problem, identify:

1. What data do I need?
2. What is the Boolean condition?
3. What happens when it is true?
4. Do I need an else?

This planning habit becomes increasingly important as selection gets more complex.



# Outline

- |                                |                                  |                                    |
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# Outline

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## 26. Practice Challenge

Write a program that asks for a number.

- If it is positive, print `Positive`.
- Otherwise print `Zero` or `negative`.



## 26. Practice Challenge

Write a program that asks for a number.

- If it is positive, print `Positive`.
- Otherwise print `Zero` or `negative`.

Extension: how could you distinguish between **zero** and **negative**?

That leads us into the next session.



# Outline

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# Outline

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**27. Session 1 Recap**



## 27. Session 1 Recap

You should now be comfortable with:

- Boolean values and conditions
- `>`, `<`, `>=`, `<=`, `==`, `!=`
- `if`
- `if-else`
- indentation
- tracing branches
- testing boundary values

Next: **more than two choices** and **combining conditions**.